## Problem 36: Convolution Integral

i. Folding
ii. Shifting
iii. Finding Area

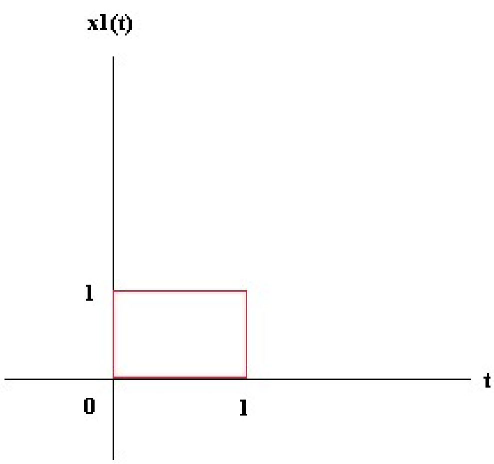
1st time:

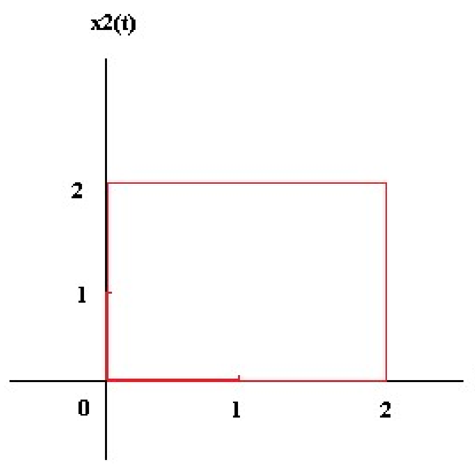
i. Folding
ii. Finding Area

2nd time and more

i. Shifting
ii. Finding Area

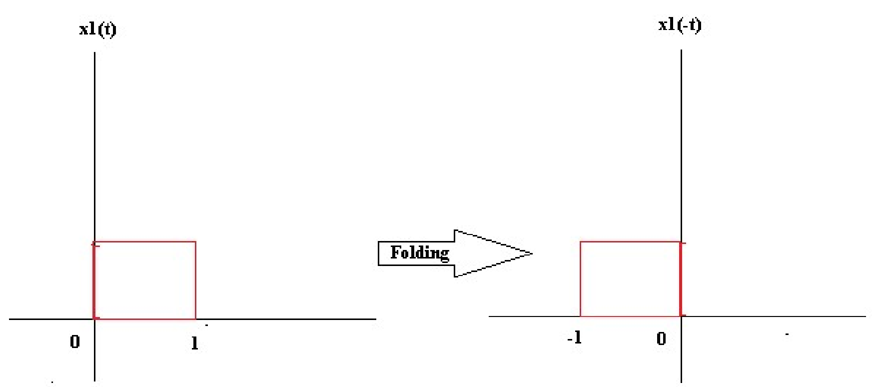
**Example 107: Find convolution integral of** $x_1(t) * x_2(t)$



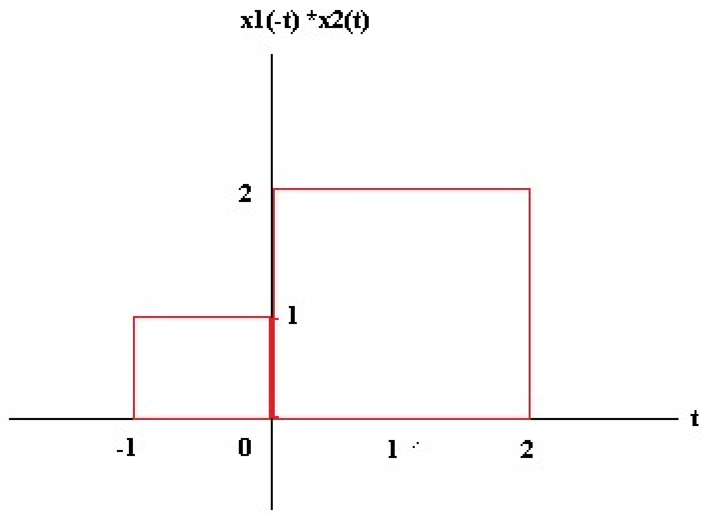




**1st time:**

i. Folding



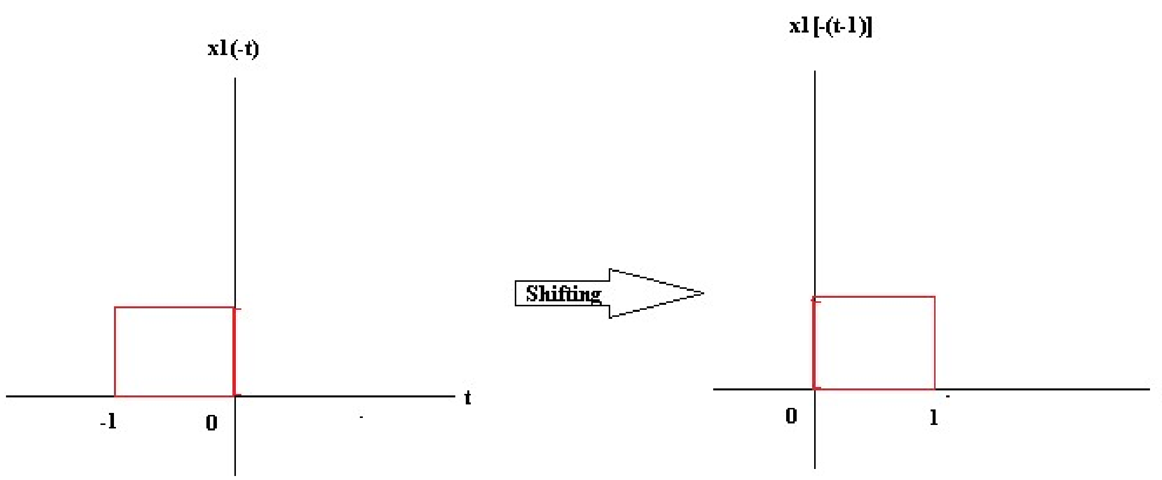
ii. Finding Area




Finding Area: No overlapped area; x[0] =0

**2$^{nd}$ time:**

i. Shifting




ii. Finding Area:



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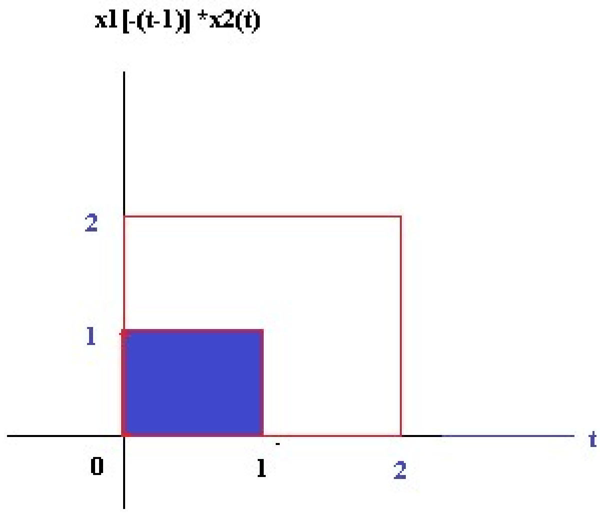
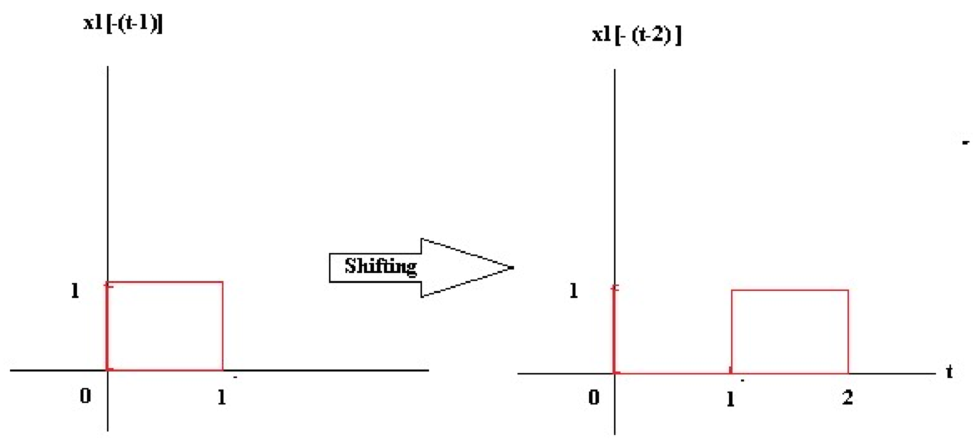
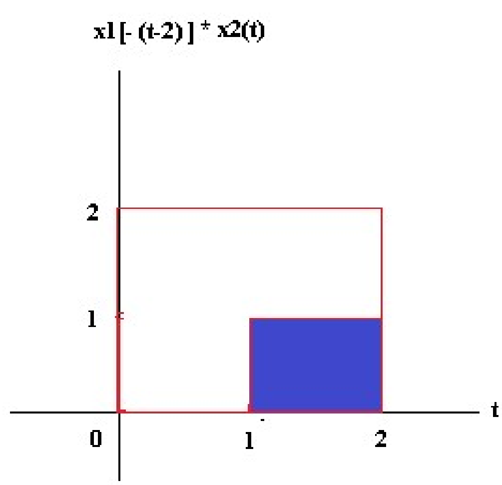



Area: x[1] = $1 \times 1 = 1$

iii. Shifting

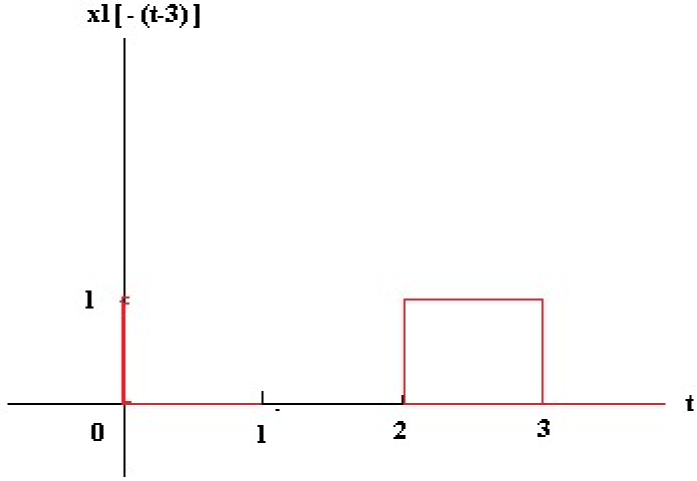



iv. Finding area

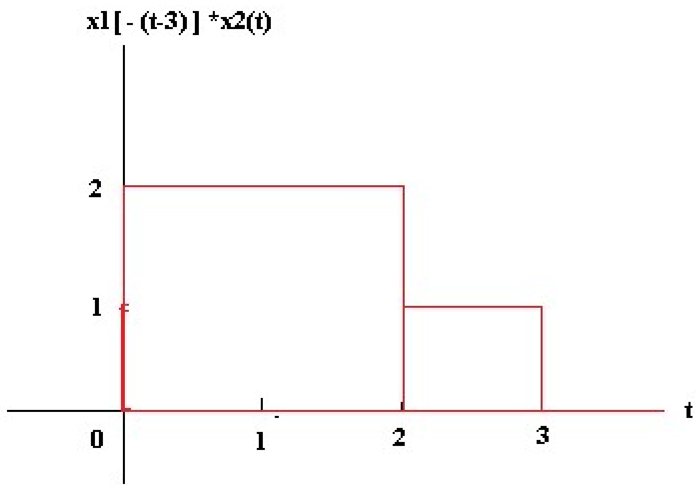


Area: x[2] = $1 \times 1 = 1$

v. Shifting:

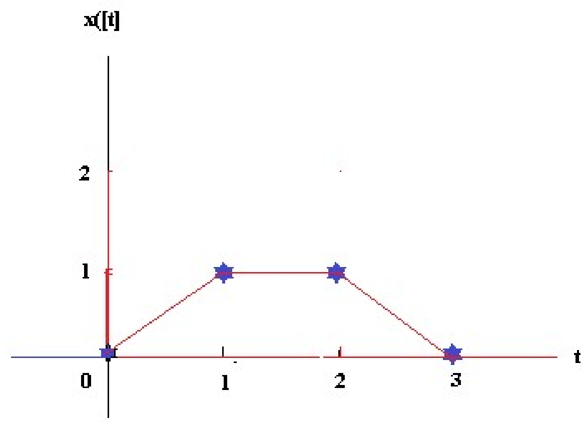



vi. Finding Area:




No overlapped: x [3] =0

Hence the convolution integral is




Home task